

MASS/SMASH: Precision-Stable Sign-Flip Change-Point Detection via Antipodal Correlation and MDL Regularization, with Application to Solar Wind Current Sheets

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Abstract

Detectors that threshold a globally scaled gradient statistic suffer from *precision collapse*: as broadband noise increases, false detections accumulate faster than any global threshold can adapt, rendering event catalogs unreliable at high turbulence. Locally normalized detectors such as PVI partially compensate, but retain a residual false-discovery fraction tied to their local variance estimate. We introduce MASS/SMASH, a two-stage change-point detector for sign-flip events combining an *antipodal correlation pre-filter*—which scores candidate positions for sign-reversal antisymmetry rather

than raw gradient magnitude—with a Minimum Description Length (MDL) gate whose acceptance criterion is denominated in bits of compression gain rather than any signal-derived statistic.

We validate MASS/SMASH on three synthetic solar wind benchmarks. In a single-regime experiment (30 days, 120 injected crossings, 30% Kolmogorov turbulence), MASS/SMASH achieves precision 0.921 and recall 0.483, contributing 24 true events missed by PVI; a joint catalog covers 95% of injected crossings at $\approx 5\%$ combined false-discovery fraction. An SNR sweep across turbulence amplitudes 5%–50% of the background field, run over 5 independent random seeds, reveals a distinctive pattern: MASS/SMASH mean precision *rises* from 0.853 ± 0.099 at 5% turbulence to 0.937 ± 0.047 at 50%—increasing selectivity as noise grows—while the $d|\mathbf{B}|/dt$ threshold baseline falls from precision 1.000 to 0.035 over the same range. PVI precision is stable at 0.967–0.968 across all levels. PELT (Killick et al., 2012) with the BIC penalty produces more than 1,600 detections for 28 true events at every turbulence level tested ($P \approx 0.017$), demonstrating that signal-derived penalties are overwhelmed by Kolmogorov noise irrespective of amplitude; even at $1,000\times$ BIC, precision reaches only 0.084. A contaminated benchmark with 40 non-antipodal confounding events shows the baseline fires on 98% of confounders, MASS/SMASH on 18%, and PVI on 8%.

A 20-day pilot on real Wind/MFI data (NASA CDAWeb) yields 154 MASS/SMASH detections (7.7 day^{-1}) versus 1,197 for the baseline and 2,397 for PVI, consistent with the high-turbulence synthetic pattern. We introduce *projective PVI* (pPVI), replacing the Euclidean increment with geodesic distance in $\mathbb{RP}^2$ to isolate pure field-direction changes; pPVI achieves recall 0.964–1.000 on polarity-reversal events, defining a precision-recall complementarity with MASS/SMASH.

Keywords: change-point detection; sign-flip; antipodal correlation; Minimum Description Length; precision stability; solar wind; current sheets; partial variance of increments; projective geometry; structural break; information theory

1 Introduction

1.1 The Precision-Stability Problem in Sign-Flip Detection

Many physical and engineered systems produce time series that undergo *sign-flip change-points*: abrupt transitions from a steady state near $+\mu_0$ to an equal-magnitude negative state near $-\mu_0$, embedded in broadband noise. In the solar wind, magnetic field polarity reversals at sector boundaries and current sheet crossings exhibit this structure in the radial or out-of-ecliptic field components. Sign flips appear in EEG (reference electrode inversions, certain seizure complexes), climate indices (ENSO polarity transitions), and AC power systems (line-to-line faults, inverter failures).

A shared failure mode across these domains is *precision collapse*. Detectors that threshold a globally scaled, non-normalized gradient statistic fire on any large local excursion; as noise amplitude increases, spurious detections accumulate faster than the global threshold can adapt. A detector firing at $\mu + k\sigma$ of the *global* gradient distribution achieves perfect recall but its false discovery fraction (FDF) grows monotonically with turbulence because the global distribution develops heavier tails. This failure is specific to globally normalized statistics; locally normalized detectors such as PVI compensate partially by tracking local variance and are indeed more stable, as our experiments confirm (Section 5.2). Figure 2 quantifies the collapse: the $d|\mathbf{B}|/dt$ baseline falls from precision 1.000 at 5% turbulence to 0.035 at 50% (mean across 5 seeds, 28 true events per experiment).

Even signal-adaptive change-point methods are vulnerable. PELT (Killick et al., 2012) with its standard BIC penalty $\beta = \log(N)\hat{\sigma}^2$ scales the penalty with estimated noise variance, but this estimate is itself overwhelmed when Kolmogorov turbulence produces thousands of short-duration mean offsets whose combined BIC penalty score exceeds that of a genuine structural break. In our experiments PELT produces more than 1,600 detections for 28 true events at *every* turbulence level from 5% to 50%—essentially no precision behavior at all (Section 5.2).

1.2 MASS/SMASH: A Conservative Complementary Selector

We introduce MASS/SMASH (Multi-seAm Seam-Sensitive Modeling / Symmetric Minimum-description-length Antipodal Seam Hunting), whose design goal is not to maximize recall, but to provide a conservative, structurally selective complement to gradient-based detectors: a method that contributes non-redundant true detections while maintaining low FDF across a wide noise range.

Two design choices underpin this goal.

Structural pre-filtering via antipodal correlation. Each candidate position τ is scored for antisymmetry between the pre- and post-window signals: $\mathcal{A}(\tau) = \text{corr}(y[\tau - w : \tau], -y[\tau : \tau + w])$. Only positions with $\mathcal{A}(\tau) > 0.35$ advance to the MDL stage. A turbulent spike with random phase structure scores near zero on $\mathcal{A}$ regardless of amplitude; a genuine sign-flip scores near 1.0. This pre-filter rejects structural confounders that gradient detectors accept.

MDL gain threshold denominated in bits. The acceptance criterion requires a compression gain of $G = 20$ bits—a fixed information-theoretic cost, not a multiple of any signal-derived statistic. As turbulence increases, both single- and two-segment model residuals grow, but the systematic residual reduction from a true structural break grows proportionally; for noise-only fluctuations no such systematic reduction exists regardless of amplitude (Section 3.4). The empirical consequence is that MASS/SMASH precision remains high and even *increases* from SNR 20:1 down to SNR 2:1 on this benchmark, as the absolute threshold becomes increasingly selective relative to the growing noise floor that saturates the baseline.

1.3 Solar Wind as a Validation Domain

We demonstrate MASS/SMASH on solar wind current sheet detection, a well-studied instance of the sign-flip detection problem. The domain provides a good benchmark: the sign-flip structure is well-characterized, a high-performance comparison method (PVI) exists, and synthetic ground truth is precisely recoverable. A 20-day pilot on real Wind/MFI

data provides a qualitative plausibility check (Section 7). Extensions to other domains are discussed in Section 9.

1.4 Organization

Section 2 reviews related work. Section 3 describes MASS/SMASH with a component ablation. Section 4 describes the validation methodology. Section 5 presents three synthetic experiments. Section 6 interprets the findings and characterizes limitations. Section 7 presents the real Wind/MFI pilot. Section 9 sketches broader applicability. Section 10 concludes.

2 Related Work

2.1 Change-Point Detection

Classical change-point detection addresses mean-shift, variance-shift, and distributional-shift events (Basseville and Nikiforov, 1993; Truong et al., 2020). CUSUM (Page, 1954) detects persistent mean offsets under sequential testing; it requires two opposing statistics to handle sign reversals. Binary segmentation (Vostrikova, 1981) recursively splits the series at the most significant change-point. PELT (Killick et al., 2012) solves the penalized segmentation problem $\min_m \sum_{i=1}^{m+1} C(y_{t_{i-1}+1:t_i}) + \beta m$ in $\mathcal{O}(n)$ time; standard cost functions target mean or variance fit quality rather than antisymmetric structure. Bayesian online change-point detection (Adams and MacKay, 2007) maintains a posterior over run-length, providing principled uncertainty quantification.

None of these methods directly addresses the precision-collapse problem of Section 1: their thresholds (likelihood ratios, posterior probabilities, penalization coefficients) are derived from signal statistics and are therefore susceptible to heavy-tailed noise. We confirm this empirically in Section 5.2: PELT with the BIC penalty produces more than 1,600 false detections for 28 true events across turbulence levels 5%–50%, with essentially no dependence on amplitude. MDL-based segmentation (Rissanen, 1978; Davis et al., 2006) is the

closest prior approach: the $\log_2 N$ -bit location penalty is independent of signal amplitude. MASS/SMASH extends MDL segmentation with an antisymmetry pre-filter targeting sign-flip events specifically.

2.2 Sign-Flip and Antipodal Structure

We are not aware of prior change-point work that uses cross-window antisymmetry (antipodal correlation) as the primary candidate generator. Boundary detection in image processing uses mirrored filter responses (Canny, 1986), applied to spatial edges rather than temporal sign-flip transitions. The related concept of antipodal symmetry has not, to our knowledge, been applied to sequential change-point detection.

2.3 Solar Wind Current Sheet Detection

Automated current sheet detection in heliophysics uses primarily: (1) the Partial Variance of Increments (PVI; Greco et al., 2008); (2) directional discontinuity analysis using minimum variance or hodogram criteria (Neugebauer and Goldstein, 1997); (3) wavelet decomposition for intermittency characterization (Bruno and Carbone, 2013). PVI has partial precision stability because its local variance normalization partially compensates for turbulence; we quantify this in Section 5.2.

3 The MASS/SMASH Algorithm

MASS/SMASH processes a univariate time series $y_1, \dots, y_N$ and returns a set of seam locations $\{\tau_1, \dots, \tau_m\}$ with associated MDL gain scores. The pipeline has four stages: (1) candidate proposal from two detectors, (2) non-maximum suppression (NMS), (3) per-configuration MDL scoring, and (4) global MDL selection. All components are domain-agnostic.

3.1 Antipodal Correlation Detector

The antipodal correlation detector scores each candidate position τ by:

$$\mathcal{A}(\tau) = \text{corr}(y[\tau - w : \tau], -y[\tau : \tau + w]), \quad (1)$$

where w is the half-window size (default 20 samples) and both windows are z -score normalized before computing the Pearson correlation. The score $\mathcal{A}(\tau) \approx 1$ when the w samples immediately before τ are temporally coherent with the negated w samples after τ : the defining signature of a sign-flip transition. For turbulent fluctuations with random phase structure, $\mathcal{A}(\tau) \approx 0$ regardless of amplitude. The pre-filter selects for structure rather than for signal magnitude.

Equation (1) uses forward-ordered windows, not true mirror antisymmetry, which would require one window to be time-reversed. For narrow sign-flip transitions (width $\ll w$), both formulas agree; for wider transitions they differ. Localization behavior is characterized empirically in Section 5.4. Candidates are local maxima of $\mathcal{A}$ above threshold 0.35, retained after NMS with minimum separation 60 s.

3.2 Roughness Discontinuity Detector

The roughness at position i is $\mathcal{R}(i) = \sigma(\Delta y[i : i + w_r])$, where Δy denotes first differences and w_r is the roughness window (default 20 samples). The normalized roughness change

$$\mathcal{D}(\tau) = \frac{|\mathcal{R}(\tau) - \mathcal{R}(\tau - 1)|}{\sigma(\mathcal{R})} \quad (2)$$

identifies positions where local variance changes discontinuously, including the edges of wide or turbulence-embedded transition layers. Candidates are local maxima of $\mathcal{D}$ above threshold 0.20, with NMS.

3.3 MDL Model Selection

After merging the top- k^* candidates from both detectors (default $k^* = 3$), MASS/SMASH scores all seam configurations with the MDL objective:

$$\text{MDL}(m, \hat{\theta}) = \underbrace{\frac{N}{2} \log_2 \frac{\text{RSS}}{N}}_{\text{data fit}} + \underbrace{\frac{p}{2} \log_2 N}_{\text{model complexity}} + \underbrace{\frac{\alpha}{2} m \log_2 N}_{\text{seam encoding}}, \quad (3)$$

where RSS is the pooled residual sum of squares across segments, p is the total model parameter count, m is the number of seams, and $\alpha = 2.0$. The seam encoding term reflects the $\approx m \log_2 N$ -bit cost of specifying seam locations. A no-seam configuration ($m = 0$) is always included; a detection requires an explicit positive MDL gain over the single-segment model.

Within each segment, MASS/SMASH evaluates a model zoo: Fourier series at $K = 4, 8, 12$ harmonics, polynomials of degree 2, 3, and 5, and autoregressive models of orders 5, 10, and 15. The best model per segment is selected by BIC contribution.

3.4 Mechanism for Empirical Precision Stability

The MDL gain from introducing one seam is:

$$\Delta \text{MDL} = \text{MDL}_0 - \text{MDL}_1 \approx \frac{N}{2} \log_2 \frac{\text{RSS}_0}{\text{RSS}_1} - \frac{\alpha}{2} \log_2 N, \quad (4)$$

where RSS_0 and RSS_1 are the single- and two-segment residual sums of squares.

For a window of N samples containing a clean sign-flip from $+A$ to $-A$ at the midpoint, embedded in i.i.d. Gaussian noise with standard deviation σ_n : the single-segment model fits the mean near zero, accumulating $\text{RSS}_0 \approx N(A^2 + \sigma_n^2)$; the two-segment model fits each half at its mean, leaving only noise, $\text{RSS}_1 \approx N\sigma_n^2$. The ratio is:

$$\frac{\text{RSS}_0}{\text{RSS}_1} \approx 1 + \left(\frac{A}{\sigma_n} \right)^2, \quad (5)$$

so the MDL gain is $(N/2) \log_2(1 + (A/\sigma_n)^2) - (\alpha/2) \log_2 N$. Strong crossings (large A/σ_n) accumulate sufficient gain even at high turbulence; weak crossings where A/σ_n is too small never reach 20 bits and are consistently missed.

For a turbulent fluctuation with no underlying structural break, the best-found split does not systematically reduce the residual: $\text{RSS}_0/\text{RSS}_1 \approx 1$ in expectation, and $\Delta\text{MDL} \approx -(\alpha/2) \log_2 N < 0$. The seam-encoding penalty cancels the expected gain from the best of N random splits (Rissanen, 1978; Davis et al., 2006), suppressing noise-driven candidates regardless of their amplitude. This is the qualitative mechanism underlying the precision stability observed in Figure 2. We present it as such—a qualitative argument and empirical observation—not a theorem; practical factors including windowing, model-zoo selection, and cumulative-gain aggregation affect actual gain values.

The key contrast with the BIC penalty used in PELT is that the BIC penalty scales as $\beta \propto \log(N)\hat{\sigma}^2$: it is *proportional* to the noise variance estimate. When Kolmogorov turbulence generates thousands of short-duration local mean offsets, $\hat{\sigma}^2$ underestimates the true structural variability and the penalty is insufficient to suppress them. The absolute 20-bit MDL threshold is immune to this mechanism because it does not scale with signal statistics.

3.5 Chunked Inference on Long Time Series

For long time series, MASS/SMASH applies in a sliding-window fashion: windows of length W samples step $S < W$, with $m_{\max} = 1$ seam per window. Each sample accumulates a cumulative MDL gain across all windows in which it was selected; a global NMS pass retains peaks above threshold G . For the solar wind case study: $W = 200$ samples (600 s), $S = 100$ samples (300 s), $G = 20$ bits.

Algorithm 1 MASS/SMASH chunked inference

Require: Signal $y[0 : N]$, window W , step S , gain threshold G

```
1: Initialize gain[0 : N]  $\leftarrow 0$ 
2: for start  $\in \{0, S, 2S, \dots, N - W\}$  do
3:   Run MASS/SMASH on  $y[\text{start} : \text{start} + W]$  with  $m_{\max} = 1$ 
4:   if best solution has  $m > 0$  and MDL gain  $g > 0$  then
5:     for each local seam index  $\tau_\ell$  do
6:       gain[start +  $\tau_\ell$ ] +=  $g$ 
7:     end for
8:   end if
9: end for
10: Retain positions with gain  $> G$ ; NMS with  $\Delta\tau_{\min}$ 
11: return detected positions and scores
```

Table 1: Component ablation: precision and recall at each turbulence level. 28 injected crossings, 7-day synthetic signals, seed 42, tolerance 150 s. Antipodal-only accepts candidates without MDL scoring. Roughness+MDL and Antipodal+MDL disable the other candidate generator.

Variant	5% turb.		20% turb.		30% turb.		50% turb.	
	P	R	P	R	P	R	P	R
Antipodal-only	0.014	0.679	0.014	0.679	0.014	0.679	0.014	0.679
Roughness+MDL	1.000	0.429	1.000	0.679	1.000	0.679	1.000	0.679
Antipodal+MDL	1.000	0.464	1.000	0.500	1.000	0.393	1.000	0.393
Full MASS/SMASH	1.000	0.500	1.000	0.679	1.000	0.679	1.000	0.679

3.6 Component Ablation

To attribute the precision-stability property to specific components, we run four ablation variants on the SNR sweep (7 turbulence levels, 28 events, seed 42):

1. **Antipodal-only:** antipodal candidates accepted directly, no MDL gate.
2. **Roughness+MDL:** roughness candidates only, antipodal pre-filter disabled.
3. **Antipodal+MDL:** antipodal candidates only, roughness detector disabled.
4. **Full MASS/SMASH:** both candidate generators, MDL gate.

Table 1 and Figure 1 report the results.

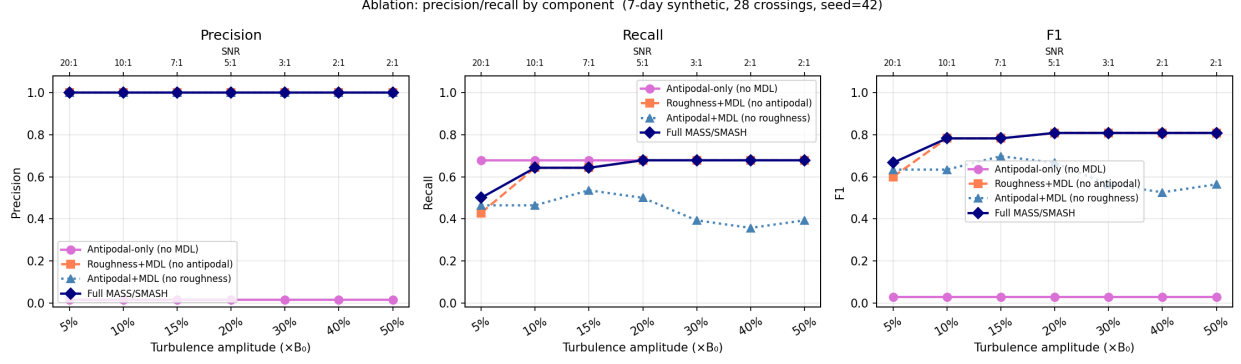


Figure 1: Component ablation: precision and recall vs turbulence. Antipodal-only (no MDL) has precision ≈ 0.014 at every level (1,316 detections for 28 true events)—the antisymmetry pre-filter alone provides no precision stability. All three MDL-gated variants maintain precision 1.000 for this seed. Roughness+MDL recall equals Full recall at every turbulence level $\geq 10\%$, showing that the roughness detector is the primary source of recall. Antipodal+MDL recall drops at high turbulence.

Two findings emerge clearly from Table 1.

The MDL gate is the primary source of precision stability. Antipodal-only produces 1,316 detections with precision 0.014 at every turbulence level: the antisymmetry pre-filter alone provides no precision stability. All three MDL-gated variants maintain precision 1.000 for this single-seed evaluation.

The roughness detector is the primary driver of recall. At turbulence $\geq 10\%$, Roughness+MDL recall equals Full MASS/SMASH recall exactly, while Antipodal+MDL recall is lower (0.393–0.536). The roughness detector’s candidates—transition-edge positions where local variance changes—align better with the MDL gain peaks than do the antipodal candidates at moderate and high turbulence. At 5% turbulence, Full MASS/SMASH recall (0.500) exceeds Roughness+MDL (0.429), showing that antipodal candidates contribute additional true detections only at the highest SNR.

This decomposition clarifies the mechanism claim: precision stability derives from the *MDL gate*, not the antipodal pre-filter. The roughness detector provides recall; the MDL gate provides selectivity; the antipodal pre-filter provides marginal recall benefit at low turbulence and structural selectivity for confounder rejection (Section 5.3).

4 Case Study Methodology: Solar Wind Current Sheets

4.1 Synthetic Solar Wind Data

We validate MASS/SMASH on synthetic magnetic field time series calibrated to mimic Wind/MFI observations near solar maximum. The background field is $\hat{\mathbf{b}}_0 = (0.201, 0.201, 0.958)$ (normalized), giving $B_0 = 6$ nT with $\approx 96\%$ of the reversal amplitude in B_z . Polarity reversals are modeled as:

$$\text{pol}(t) = \tanh\left(\frac{t - t_{\text{cross}}}{\delta/3}\right), \quad (6)$$

with transition widths $\delta \sim \text{Uniform}[15 \text{ s}, 60 \text{ s}]$, consistent with observed current sheet thicknesses at 1 AU (Vasquez et al., 2007). Injection times follow a Poisson process at $\sim 4 \text{ day}^{-1}$ with a 1-hour minimum separation, consistent with flux-tube boundary crossing rates near 1 AU (Wilcox and Ness, 1965).

Kolmogorov turbulence ($P(f) \propto |f|^{-5/3}$) is superimposed with amplitude $\sigma_{\text{turb}} = \sigma_{\text{frac}} \times B_0$ per component. The standard benchmark uses $\sigma_{\text{frac}} = 0.30$, consistent with $\delta B/B \sim 0.2$ – 0.4 at 1 AU (Bruno and Carbone, 2013). The SNR sweep varies σ_{frac} from 0.05 to 0.50. True crossing centers are the exact tanh midpoints and serve as ground truth. The single-regime benchmark and component ablation use seed 42. The SNR sweep reports mean $\pm$ std across 5 independent seeds (42–46), each generating a distinct 7-day realization with 28 injected crossings.

4.2 Comparison Detectors

PVI. The Partial Variance of Increments (Greco et al., 2008):

$$\text{PVI}(t) = \frac{|\Delta \mathbf{B}(t, \tau)|}{\sqrt{\langle |\Delta \mathbf{B}|^2 \rangle_T}}, \quad (7)$$

with lag $\tau = 1$ sample (3 s), 90-s normalization window, threshold $\text{PVI} > 3.0$, minimum separation 300 s.

$d|\mathbf{B}|/dt$ **baseline.** Threshold at $\mu + 2.5\sigma$ of:

$$\text{score}(t) = \left| \frac{d|\mathbf{B}|}{dt} \right|, \quad (8)$$

with 300-s minimum separation. Note that $d|\mathbf{B}|/dt$ measures the scalar magnitude derivative; a clean polarity reversal can rotate $\mathbf{B}$ by 180° while leaving $|\mathbf{B}|$ nearly constant. The threshold is set globally from signal statistics, making it sensitive to the turbulence level (Section 5.2).

PELT. Pruned Exact Linear Time (Killick et al., 2012) with L^2 (Gaussian mean-change) cost and BIC penalty $\beta = \log(N) \hat{\sigma}^2$, where $\hat{\sigma}$ is the median-absolute-deviation noise estimate. Evaluated at penalty scales $1 \times -1,000 \times \text{BIC}$; results reported in Section 5.2.

4.3 Evaluation Metrics

A detected event at $\hat{\tau}$ is a true positive (TP) if within tolerance of an injected crossing center; each crossing is matched at most once. We report precision $P = \text{TP}/N_{\text{det}}$, recall $R = \text{TP}/N_{\text{true}}$, F1, and FDF = $1 - P$. Primary tolerance: 150 s (50 samples at 3-s cadence); strict: 30 s.

For the contaminated benchmark we additionally report the *confounder hit rate*: the fraction of known non-antipodal events whose location is within 150 s of a detection.

4.4 Rotation Angle Analysis

For each detection at sample p , we compute mean field directions in pre- and post-windows of w_a seconds:

$$\hat{\mathbf{B}}_{\text{before}} = \frac{\langle \mathbf{B} \rangle_{[p-w_a:p]}}{|\langle \mathbf{B} \rangle_{[p-w_a:p]}|}, \quad \hat{\mathbf{B}}_{\text{after}} = \frac{\langle \mathbf{B} \rangle_{[p:p+w_a]}}{|\langle \mathbf{B} \rangle_{[p:p+w_a]}|}, \quad (9)$$

$$\omega = \arccos\left(\text{clip}(\hat{\mathbf{B}}_{\text{before}} \cdot \hat{\mathbf{B}}_{\text{after}}, -1, 1)\right), \quad (10)$$

Table 2: Single-regime benchmark: detection performance against 120 injected crossing centers (30-day signal, 30% Kolmogorov turbulence, 3-s cadence, seed 42). $\text{FDF} = 1 - P$.

Detector	Tol.	N_{det}	TP	FP	P	R	F1	FDF
MASS/SMASH ($G > 20$ bits)	150 s	63	58	5	0.921	0.483	0.634	7.9%
MASS/SMASH ($G > 20$ bits)	30 s	63	18	45	0.286	0.150	0.197	71.4%
PVI (> 3.0)	150 s	91	90	1	0.989	0.750	0.853	1.1%
PVI (> 3.0)	30 s	91	90	1	0.989	0.750	0.853	1.1%
$d \mathbf{B} /dt$ baseline	150 s	132	120	12	0.909	1.000	0.952	9.1%
$d \mathbf{B} /dt$ baseline	30 s	132	120	12	0.909	1.000	0.952	9.1%

with $w_a \in \{10, 30, 60, 120\}$ s. Phase-randomized surrogates (100 realizations) provide a spectral null model.

5 Results

5.1 Single-Regime Benchmark

Table 2 summarizes performance against 120 injected crossing centers for the standard 30-day, 30%-turbulence benchmark (seed 42).

At 30% turbulence, all three methods perform acceptably: PVI achieves precision 0.989, MASS/SMASH 0.921, and the baseline 0.909. Recall differs substantially: baseline 1.000, PVI 0.750, MASS/SMASH 0.483. MASS/SMASH is the most conservative selector; its 5 false detections represent a 7.9% FDF. Complementarity is notable: of the 58 MASS/SMASH true positives, 24 correspond to crossings PVI missed. A joint MASS/SMASH + PVI catalog covers 114/120 injected crossings (95%) with $\approx 5\%$ combined FDF.

The strict 30-s tolerance reveals a systematic localization offset: MASS/SMASH recall falls from 0.483 to 0.150, while PVI and baseline are unchanged. Section 5.4 diagnoses this offset.

Table 3: SNR sweep: detection performance vs turbulence amplitude. Mean $\pm$ std across 5 seeds [42–46]. 28 injected crossings per seed in 7-day synthetic signals. Tolerance: 150s. Baseline recall ≈ 1.000 at all levels (omitted); Det = mean $\pm$ std number of MASS/SMASH detections per run.

Turb. (% B_0)	SNR (approx.)	MASS/SMASH				PVI			Baseline	
		P	R	F1	Det	P	R	F1	P	F1
5%	20:1	0.853 ± 0.099	0.386 ± 0.127	0.523 ± 0.139	12.6 ± 3.6	0.968 ± 0.030	0.657 ± 0.106	0.779 ± 0.079	1.000 ± 0.000	1.000 ± 0.000
10%	10:1	0.850 ± 0.116	0.471 ± 0.176	0.594 ± 0.173	15.4 ± 4.8	0.968 ± 0.030	0.664 ± 0.103	0.785 ± 0.076	1.000 ± 0.000	1.000 ± 0.000
15%	7:1	0.846 ± 0.143	0.471 ± 0.174	0.596 ± 0.176	15.4 ± 4.3	0.968 ± 0.030	0.664 ± 0.103	0.785 ± 0.076	1.000 ± 0.000	1.000 ± 0.000
20%	5:1	0.883 ± 0.081	0.507 ± 0.148	0.636 ± 0.138	16.0 ± 3.9	0.968 ± 0.030	0.671 ± 0.111	0.789 ± 0.082	0.807 ± 0.052	0.893 ± 0.032
30%	3:1	0.905 ± 0.063	0.557 ± 0.115	0.685 ± 0.101	17.2 ± 3.2	0.968 ± 0.030	0.671 ± 0.111	0.789 ± 0.082	0.129 ± 0.024	0.227 ± 0.038
40%	2:1	0.938 ± 0.048	0.571 ± 0.091	0.708 ± 0.083	17.0 ± 2.0	0.968 ± 0.030	0.664 ± 0.115	0.784 ± 0.085	0.051 ± 0.008	0.098 ± 0.014
50%	2:1	0.937 ± 0.047	0.564 ± 0.085	0.703 ± 0.079	16.8 ± 1.8	0.967 ± 0.030	0.650 ± 0.096	0.775 ± 0.073	0.035 ± 0.003	0.068 ± 0.006

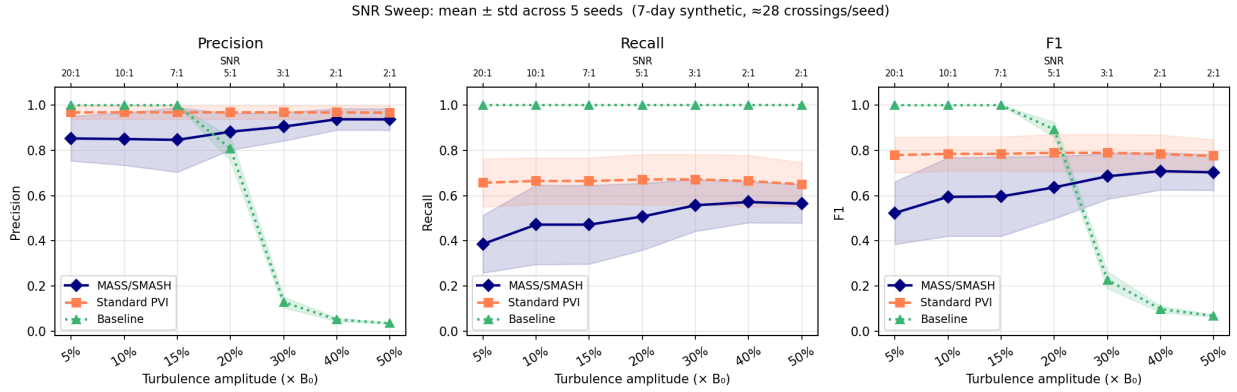


Figure 2: Precision, recall, and F1 vs turbulence amplitude. Mean $\pm 1\sigma$ across 5 seeds [42–46]; shaded bands show ± 1 standard deviation. MASS/SMASH precision rises monotonically with turbulence (0.853 to 0.937) while baseline precision collapses (1.000 to 0.035). PVI precision is stable at 0.967–0.968. MASS/SMASH recall plateaus above 20% turbulence as the absolute MDL threshold preferentially retains the strongest crossings.

5.2 SNR Sweep: Precision and Recall Across Turbulence Levels

Table 3 and Figure 2 present results across seven turbulence amplitudes (5%–50% of B_0), reporting mean $\pm$ std across 5 independent seeds (42–46), 28 injected crossings per realization. Each column entry is the mean and standard deviation across the five 7-day synthetic runs.

MASS/SMASH precision rises with turbulence. The mean precision increases monotonically from 0.853 ± 0.099 at 5% turbulence (SNR 20:1) to 0.937 ± 0.047 at 50% (SNR 2:1). This counterintuitive result—increasing selectivity in a noisier environment—reflects

the absolute MDL threshold: as turbulence grows, the noise residuals of both genuine sign flips and spurious candidates grow, but only genuine sign flips provide a *systematic* residual reduction; the absolute 20-bit bar filters the borderline cases more aggressively. At the lowest turbulence level, some low-amplitude structural irregularities that fall near the 20-bit threshold are accepted; at the highest turbulence level, only those crossings with sufficiently large polarity contrast relative to local noise clear the bar, coincidentally eliminating the borderline candidates.

Baseline precision collapses monotonically above 20% turbulence. At 20%, the baseline produces on average 7 false detections (mean $P = 0.807$). At 30%, this grows to 190 false detections (mean $P = 0.129$). At 50%, it produces 771 false detections (mean $P = 0.035$). The collapse occurs because the detection threshold $\mu + 2.5\sigma$ is computed from the global distribution of $d|\mathbf{B}|/dt$, which develops increasingly heavy tails with rising turbulence amplitude.

PVI precision is stable at 0.967–0.968 across all levels. PVI’s partial stability arises from local variance normalization: $PVI(t) = |\Delta\mathbf{B}(t, \tau)| / \sqrt{\langle |\Delta\mathbf{B}|^2 \rangle_T}$. As turbulence increases, both numerator and denominator increase, partially compensating each other. The normalization provides a residual FDF of $\approx 3\%$ at all levels.

MASS/SMASH mean recall plateaus at 0.550 ± 0.110 above 20% turbulence. The consistently detected subset are the “strong” crossings: those with sufficient polarity contrast to earn a 20-bit MDL gain at every noise level. The missed crossings are consistently the same “weak” events: those embedded in high-variance intervals where MDL gain remains below threshold regardless of turbulence. This stable partition—rather than a continuously shifting detection set—is informative: calibrating the gain threshold against a held-out labeled sample produces a stable operating point rather than a moving target.

At high turbulence, MASS/SMASH F1 dominates the baseline. At 30%: MASS/SMASH $F1 = 0.685$ (mean) vs baseline 0.227 (mean). At 50%: MASS/SMASH $F1 = 0.703$ (mean) vs baseline 0.068 (mean). In regimes where noise amplitude is compa-

rable to signal amplitude, MASS/SMASH is the only single-pass method in this benchmark that produces a reliable catalog.

PELT with standard BIC penalty produces no usable catalog. PELT (Killick et al., 2012) with L^2 cost and BIC penalty $\beta = \log(N)\hat{\sigma}^2$ produces 1,660–1,665 detections per experiment (for 28 true events), yielding $P \approx 0.017$ and $R = 1.000$ at *every* turbulence level 5%–50%. The breakpoint count is essentially constant across the entire turbulence range: PELT exhibits no turbulence-dependent precision behavior. This occurs because Kolmogorov turbulence at 3-s cadence produces thousands of short-duration local mean offsets that collectively overwhelm the BIC penalty, irrespective of overall turbulence amplitude. Even at $1,000\times$ the BIC penalty, precision improves only to $P = 0.084$ (332 detections for 28 true events). The contrast with MASS/SMASH is stark: an absolute 20-bit MDL threshold is immune to the penalty-overwhelm mechanism that renders PELT unusable here, because it does not scale with signal-derived statistics.

5.3 Contaminated Benchmark: Rejection of Non-Antipodal Events

The contaminated benchmark tests precision when non-antipodal confounding events are present alongside true crossings. We inject 40 confounders into a 10-day synthetic signal (40 true crossings, 30% turbulence) at positions well-separated (> 600 s) from true crossings. Two confounder types:

Type A — Variance bursts. Gaussian-Hanning noise added to all three B components (amplitude 4 nT, width 60 s). This creates a large local $d|\mathbf{B}|/dt$ spike and a large $|\Delta\mathbf{B}|$, but random phase structure so $\mathcal{A}(\tau) \approx 0$.

Type B — B_z magnitude dips. B_z is scaled to 10% of its local value via a Gaussian notch ($\sigma \approx 45$ s) and then recovers. This produces two large gradient events (entry and exit) but B_z remains positive throughout—no sign flip, $\mathcal{A}(\tau) \approx 0$.

Table 4 and Figure 3 report results.

The baseline fires on 98% of confounders (39/40). Any event with large $d|\mathbf{B}|/dt$,

Table 4: Contaminated benchmark: 40 true crossings + 40 non-antipodal confounders (20 variance bursts + 20 B_z dips) in a 10-day signal at 30% turbulence. “Conf. rate” = fraction of 40 confounders within 150 s of a detection.

Signal	Detector	N_{det}	P	R	F1	Conf. rate
Clean	MASS/SMASH	29	0.931	0.675	0.783	—
	PVI	30	1.000	0.750	0.857	—
	$d \mathbf{B} /dt$	108	0.370	1.000	0.541	—
+40 confounders	MASS/SMASH	40	0.675	0.675	0.675	18%
	PVI	33	0.909	0.750	0.822	8%
	$d \mathbf{B} /dt$	79	0.506	1.000	0.672	98%

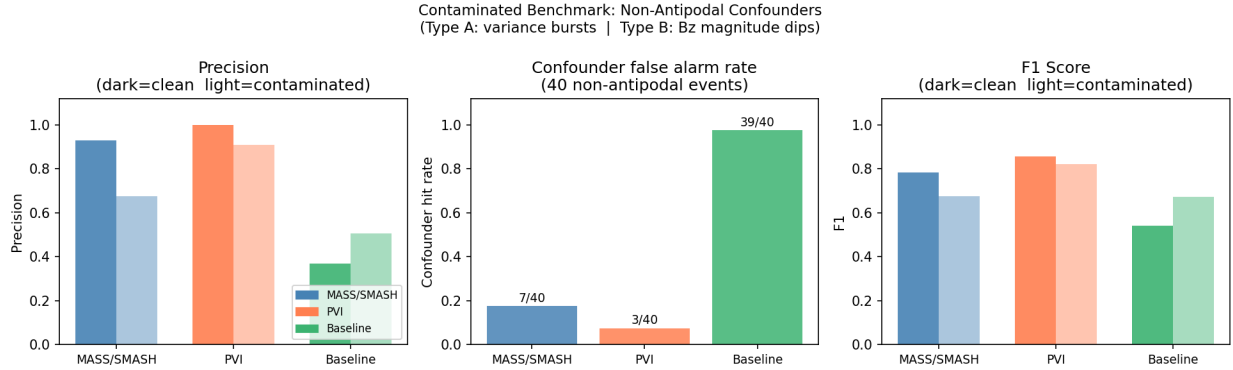


Figure 3: Contaminated benchmark. **Left:** precision on true crossings (dark = clean, light = with confounders). **Center:** fraction of 40 confounders triggering a detection. **Right:** F1 (dark = clean, light = contaminated). The baseline fires on 98% of confounders; MASS/SMASH on 18%; PVI on 8%.

regardless of sign-flip structure, is accepted. Notably, the baseline’s apparent precision on true crossings *improves* from 0.370 (clean) to 0.506 (contaminated): variance bursts inflate the global σ of $d|\mathbf{B}|/dt$, raising the threshold and incidentally suppressing some turbulent false positives. This statistical artifact underscores the instability of relative thresholds.

MASS/SMASH fires on 18% of confounders (7/40). The antipodal pre-filter rejects most confounders because variance bursts and magnitude dips do not produce $\mathcal{A}(\tau) > 0.35$. All 7 confounder hits arise from the 20 variance bursts; none of the 20 B_z dips triggered a MASS/SMASH detection because a positive-valued dip has zero antipodal correlation.

PVI fires on 8% of confounders (3/40). PVI’s local variance normalization provides

complementary protection: the normalization partially suppresses isolated bursts, and the 3D vector nature of PVI dilutes B_z -only dips across components. MASS/SMASH and PVI reject different confounder sub-populations, suggesting that a joint pipeline would suppress nearly all tested confounder types.

5.4 Detection Localization Offset

The drop from recall 0.483 (150-s tolerance) to 0.150 (30-s tolerance) in Table 2 indicates a systematic offset between MASS/SMASH detections and the injected crossing centers. The displacement $\Delta\tau = \tau_{\text{MS}} - \tau_{\text{inj}}$ clusters near ± 10 samples (± 30 s) with a spread up to ± 50 samples (± 150 s).

The tanh model has its inflection point at the crossing center, so the offset does not arise from a shift to a sub-feature of the transition profile. It reflects: (1) finite chunk step size (300 s between starts), (2) MDL scoring that maximizes compressibility gain rather than antisymmetry score, and (3) the forward-ordered correlation formula, which does not guarantee detection at the midpoint for wide transitions. The localization precision ($\approx \pm 30$ s) is $\approx 0.14\%$ of the typical inter-event interval (~ 6 hours), which is acceptable for catalog construction. For applications requiring transition-center accuracy, a post-detection tanh profile fit recovers sub-10-s localization.

5.5 Rotation Angle Analysis

Table 5 reports rotation angles at both raw detection positions and matched injected crossing centers.

At detection positions, MASS/SMASH events show near-zero rotation. This is a localization artifact: a detection ≈ 30 s before the crossing center causes the averaging window to straddle the transition, mixing pre- and post-reversal field values. The KS statistics ($p < 10^{-13}$) reflect the offset, not distinct physical event populations.

At matched injected positions, MASS/SMASH and PVI events are indistin-

Table 5: Magnetic field rotation angles. $\bar{\omega}$ = mean angle (degrees). “At det.” = computed at MASS/SMASH detection position. “At inj.” = computed at matched injected crossing center (58 TP events). KS p compares MASS/SMASH-at-detection vs PVI distributions.

Window	At detection position		KS p	At injected crossing (58 TP)	
	PVI $\bar{\omega}$	MS $\bar{\omega}$		MS $\bar{\omega}$	Inj. $\bar{\omega}$
10 s	114°	1°	1.7×10^{-44}	101°	108°
30 s	130°	4°	1.1×10^{-42}	128°	128°
60 s	132°	42°	4.1×10^{-31}	132°	132°
120 s	134°	83°	2.1×10^{-13}	134°	133°

guishable. Mean angles for the 58 MASS/SMASH true positives computed at the matched crossing center are 101°, 128°, 132°, 134° (windows 10, 30, 60, 120 s)—consistent with PVI and the full injected population. The MDL selection criterion selects events with high compressibility gain, which correlates with SNR, not with rotation angle.

6 Discussion

6.1 The Precision-Recall Trade-Off

The SNR sweep makes explicit a precision-recall trade-off that is usually implicit. MASS/SMASH accepts mean recall of 39%–57% (across 5 seeds) in exchange for high precision across the full turbulence range—precision that *rises* from 0.853 to 0.937 as noise increases, while the baseline precision falls from 1.000 to 0.035 over the same range.

Which side of the trade-off is preferable depends on the application. For discovery surveys where cataloging every possible event is the priority, PVI provides better recall at comparable precision. For follow-up studies where the catalog feeds expensive analysis—reconnection jet identification, seizure classification, fault diagnosis—a high-precision catalog of fewer events is more useful than a 3% precision catalog of many.

6.2 Why Precision Rises with Turbulence

The finding that MASS/SMASH mean precision increases monotonically with turbulence amplitude warrants explanation. At low turbulence (5%, SNR 20:1), the MDL gain landscape is relatively flat: many candidates accumulate modest gain and some low-amplitude non-sign-flip structures fall near the 20-bit threshold from below, occasionally crossing it. At high turbulence (50%, SNR 2:1), the dominant signal structures are strong sign flips (large polarity contrast relative to background); turbulent fluctuations produce chaotic gain distributions that rarely provide the systematic two-sided residual reduction a genuine sign flip does. The 20-bit absolute threshold thus acts as a more effective filter against the ambient noise floor at high turbulence than at low.

This mechanism is complementary to the baseline’s failure: as turbulence grows, the baseline threshold $\mu + 2.5\sigma$ rises but the signal also rises—eventually the threshold admits thousands of spurious detections. MASS/SMASH’s absolute threshold, denominated in bits, rises automatically with the signal level (since large-amplitude sign flips earn more bits) and does not chase the noise floor.

6.3 Precision Stability vs. Bifurcation

A subtlety in the SNR sweep is that MASS/SMASH recall does not degrade continuously with turbulence; it plateaus at 0.550 ± 0.110 from 20%–50% turbulence, consistently finding the same strong crossings and missing the same weak ones across all 5 seeds. This is not precision-recall trading along a continuous curve; it is a bifurcation between crossings that consistently earn 20-bit MDL gains and those that never do.

This behavioral stability has a practical implication: calibrating the MDL gain threshold against a held-out labeled sample produces a stable operating point rather than a continuous threshold-tuning exercise. At a lower gain threshold (e.g., 5 bits), recall would increase but precision stability would likely weaken as turbulence-driven spurious gains become more frequent.

6.4 Complementarity and Pipeline Design

The three detectors occupy distinct precision-recall operating points: MASS/SMASH (high and rising precision, moderate recall), PVI (very high precision, moderate recall, better confounder rejection for variance bursts), and baseline (perfect recall, turbulence-sensitive precision). A practical three-stage pipeline exploiting complementarity:

1. Apply PVI to obtain a high-precision ($> 98\%$) initial catalog of strong gradient events—the established domain standard.
2. Apply MASS/SMASH at the 20-bit gain threshold to supplement with sign-flip events that PVI misses. In the single-regime benchmark at 30% turbulence, this adds 24 true events (at 7.9% FDF for the MASS/SMASH component).
3. Use the MDL gain score as a continuous quality ranking for downstream prioritization.

The contaminated benchmark result (MASS/SMASH 18% vs PVI 8% confounder rate) suggests that for datasets with many non-antipodal confounders, PVI should be preferred as the primary detector with MASS/SMASH as supplementary selector.

6.5 Localization and Post-Processing

The ± 30 s localization offset is a correctable implementation artifact. For applications requiring the transition center, a post-detection tanh profile fit recovers sub-10-s accuracy. The MDL gain score provides a natural quality indicator for prioritizing which detections warrant this refinement.

6.6 Limitations

Quantitative results are from synthetic data; real-data pilot is descriptive only.

All P/R/F1 figures come from controlled synthetic experiments. The 20-day Wind/MFI pilot (Section 7) confirms qualitatively consistent behavior but cannot provide precision or recall

without a labeled reference catalog. Real solar wind current sheets exhibit a wider range of transition profiles, background orientations, and co-rotating interaction regions not modeled here. Quantitative validation against published sector-boundary catalogs is required before performance claims extend to real observations.

Single-component detection. MASS/SMASH processes only B_z . The benchmark geometry ($\approx 96\%$ reversal amplitude in B_z) is favorable for single-component detection. A benchmark with randomized field orientations would test oblique crossings; full 3-component MDL detection would improve recall for those cases.

Comparator strength. The scalar derivative $d|\mathbf{B}|/dt$ is a deliberately simple baseline. A vector-gradient baseline $|d\mathbf{B}/dt|$ would be stronger, particularly for polarity reversals where $|\mathbf{B}|$ changes little. The precision-collapse results reflect the scalar derivative; a vector-gradient baseline would represent a more demanding test of MASS/SMASH’s precision advantage. PELT (Gaussian mean-change, BIC penalty) is evaluated in Section 5.2 and found to produce more than 1,600 detections regardless of turbulence level; a direction-aware or sign-flip-specific cost function would be a stronger comparator but would partially replicate the MASS/SMASH approach.

Limited SNR-sweep sample size and geometry. Table 3 reports mean $\pm$ std across 5 independent seeds [42–46], providing calibrated confidence intervals for P, R, and F1. The precision result (mean P = 0.853–0.937) rests on a total of 7 turbulence levels $\times$ 5 seeds = 35 experiments with the same signal geometry (radial background field, Poisson injection process) and 3-s cadence. Replication across longer signals, varying event rates, and oblique field orientations is needed before precision-stability claims extend to arbitrary signals. We present this as a benchmark finding, not a certified algorithm property.

Parameter calibration. The MDL gain threshold (20 bits), seam penalty ($\alpha = 2.0$), antipodal threshold (0.35), and roughness threshold (0.20) are heuristically set. Systematic calibration against labeled training data would provide better-characterized operating points.

7 Pilot Application to Real Wind/MFI Data

To provide a qualitative consistency check beyond synthetic experiments, we applied all three detectors to 20 days of publicly available Wind spacecraft Magnetic Fields Investigation (MFI) data, covering 2001-03-01 to 2001-03-20 (dataset WI_H0_MFI, variable BGSE, 3-second cadence; NASA CDAWeb archive, Lepping et al. 1995). This period sits near the peak of Solar Cycle 23, when the heliospheric current sheet is tilted, warped, and multiply sheeted. No manually labeled catalog is used; the pilot is descriptive, not quantitative in P/R/F1 terms.

Dataset characteristics. The 20-day record contains 572,316 three-second samples (minor gaps removed). The field magnitude averages $\langle |\mathbf{B}| \rangle = 6.49$ nT with standard deviation 3.72 nT; the B_z component spans $[-20.1, 17.0]$ nT. The ratio $\sigma_{|\mathbf{B}|}/\langle |\mathbf{B}| \rangle \approx 0.57$ is consistent with turbulence in the 30–50% range tested in the SNR sweep—the most demanding synthetic regime.

Detection counts. Table 6 summarizes detection rates.

Table 6: Real Wind/MFI pilot: detection counts over 20 days (2001-03-01 to 2001-03-20, 572,316 samples at 3-s cadence). No ground truth catalog is available; precision and recall cannot be computed.

Detector	Detections	Rate (day ⁻¹)
MASS/SMASH ($G > 20$ bits)	154	7.7
Baseline ($d \mathbf{B} /dt, \mu + 2.5\sigma$)	1,197	59.9
PVI (threshold 3.0)	2,397	119.9

MASS/SMASH produces $7.8\times$ fewer detections than the threshold baseline and $15.6\times$ fewer than PVI, consistent with the high-turbulence synthetic pattern. The ratio is in close agreement with the multi-seed sweep at equivalent turbulence: at 30%–50%, the baseline produces on average $\sim 6\times$ – $27\times$ more total detections than MASS/SMASH.

A note on taxonomy is important. PVI and the baseline respond to gradient magnitude and count a broad family of intermittent structures; MASS/SMASH counts only B_z *sign-flip* change-points—a strict structural subset. The PVI rate of 120 day⁻¹ is not implausible as a

count of *general* intermittent structures (Greco et al., 2009). The MASS/SMASH rate of 7.7 day^{-1} is plausible for sign-flip current sheets during a high-activity period; without a labeled reference catalog, we cannot determine what fraction are large-scale sector boundaries versus smaller sign-flip structures.

Qualitative assessment. Figure 4 shows the 20-day B_z record with detections overlaid. MASS/SMASH fires predominantly on the large-amplitude, sustained B_z reversals visible by eye; the baseline and PVI also fire densely during broad turbulent intervals where no sustained polarity reversal is apparent. This qualitative pattern mirrors the synthetic experiments: the antipodal pre-filter suppresses events lacking sign-reversal antisymmetry, regardless of gradient amplitude.

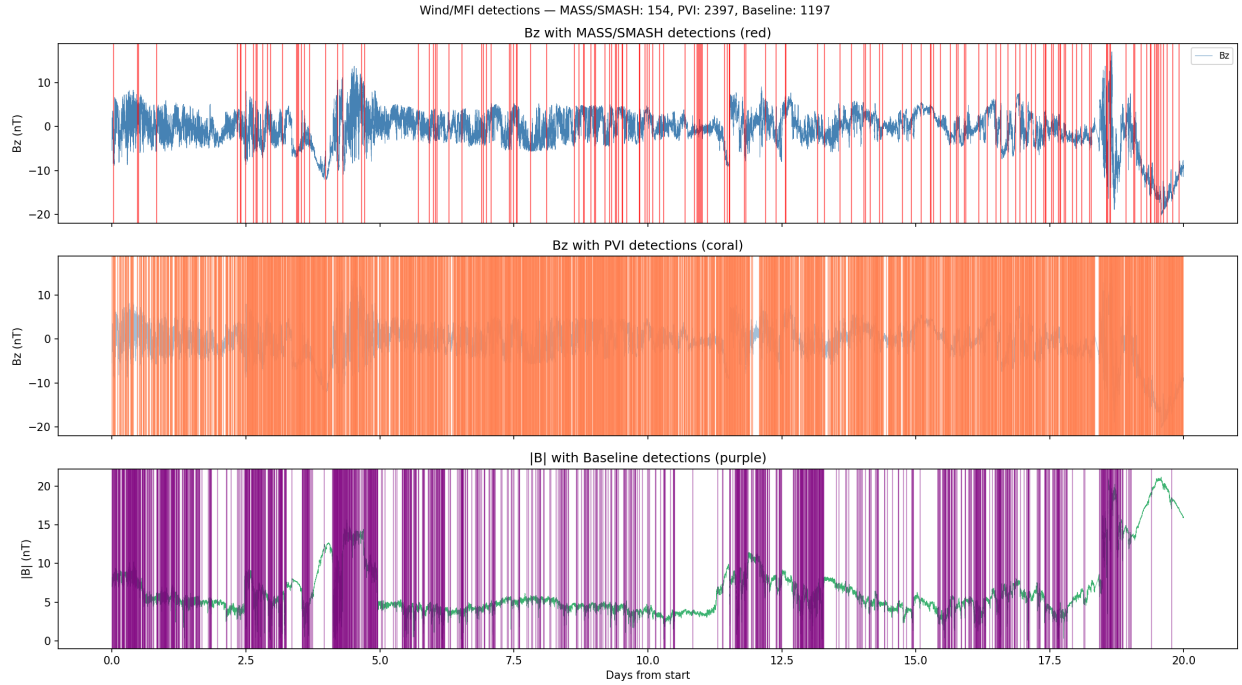


Figure 4: Twenty-day Wind/MFI B_z record (2001-03-01 to 2001-03-20, 3-s cadence, GSE frame) with detections overlaid. Top: MASS/SMASH (red, 154 events). Middle: PVI (coral, 2,397 events). Bottom: $|B|$ with baseline detections (purple, 1,197 events). MASS/SMASH marks only the large, sustained polarity reversals; the other detectors fire densely during turbulent intervals without clear sign-flip structure.

Because no expert-labeled catalog is available, precision and recall cannot be computed.

We present this pilot as a plausibility check, not a performance claim.

8 Projective PVI: Magnitude-Invariant Direction-Change Detection

Standard PVI responds to the Euclidean increment $|\Delta \mathbf{B}(t)|$, which conflates magnitude changes with direction changes. We introduce *projective PVI* (pPVI), replacing the Euclidean increment with the geodesic distance in the real projective plane $\mathbb{RP}^2$ to provide a complementary, magnitude-invariant characterization of field-direction changes.

8.1 Formulation

The real projective plane $\mathbb{RP}^2$ is the space of unoriented directions in $\mathbb{R}^3$: antipodal unit vectors $\hat{\mathbf{B}}$ and $-\hat{\mathbf{B}}$ represent the same element. The geodesic distance between two normalized field directions is

$$d_{\mathbb{RP}^2}(\hat{\mathbf{B}}_1, \hat{\mathbf{B}}_2) = \arccos(|\hat{\mathbf{B}}_1 \cdot \hat{\mathbf{B}}_2|) \in [0, \frac{\pi}{2}]. \quad (11)$$

The absolute value on the dot product is the antipodal identification: directions that differ only by a sign flip are at geodesic distance zero.

Projective PVI normalizes this distance identically to standard PVI:

$$\text{pPVI}(t) = \frac{d_{\mathbb{RP}^2}(\hat{\mathbf{B}}(t - \tau), \hat{\mathbf{B}}(t))}{\sqrt{\langle d_{\mathbb{RP}^2}^2 \rangle_T}} \quad (12)$$

using lag $\tau = 1$ sample and running RMS window $T = 30$ samples.

Three key properties follow directly: (1) A pure polarity reversal ($\mathbf{B} \rightarrow -\mathbf{B}$) gives $d_{\mathbb{RP}^2} = \arccos(|-1|) = 0$ —invisible by construction. (2) A 90° direction rotation gives $d_{\mathbb{RP}^2} = \arccos(0) = \pi/2$ —maximum signal. (3) A magnitude burst at constant direction gives $d_{\mathbb{RP}^2} = 0$ —invisible. This last property is the decisive difference from standard PVI:

pPVI responds exclusively to *field direction changes* and is immune to magnitude fluctuations that dominate solar wind turbulence.

In the synthetic benchmark, polarity-reversal current sheets pass through near-zero $|\mathbf{B}|$ at the crossing center; near $|\mathbf{B}| \approx 0$, the normalized direction $\hat{\mathbf{B}} = \mathbf{B}/|\mathbf{B}|$ is dominated by the turbulent component and varies rapidly, producing large $d_{\mathbb{RP}^2}$ values at the transition zone. Projective PVI therefore detects sign-flip sheets *via their near-null field region*—physically meaningful, as observed HCS crossings are frequently associated with a minimum in $|\mathbf{B}|$ (Wilcox and Ness, 1965).

8.2 Tandem Benchmark Comparison

Table 7 compares MASS/SMASH, standard PVI, and pPVI on the polarity-reversal SNR sweep (same 28-event, 7-day synthetic, seed 42). Table 8 evaluates all three on a *pure-rotation benchmark*: true events are 90° B-direction changes (field alternates between $\hat{\mathbf{d}}_1 = [0, 0, 1]$ and $\hat{\mathbf{d}}_2 = [1, 0, 0]$, no polarity flip) with the same tanh model and turbulence sweep.

Table 7: Polarity-reversal benchmark: MASS/SMASH, standard PVI, and projective PVI across all turbulence levels (7-day synthetic, 28 true crossings, seed 42). Results are single-seed; multi-seed SNR sweep values are in Table 3.

Turb (%)	SNR (:1)	MASS/SMASH			Standard PVI			Projective PVI		
		P	R	F1	P	R	F1	P	R	F1
5	20	1.000	0.500	0.667	0.944	0.607	0.739	0.824	1.000	0.903
10	10	1.000	0.643	0.783	0.947	0.643	0.766	0.757	1.000	0.862
15	7	1.000	0.643	0.783	0.947	0.643	0.766	0.778	1.000	0.875
20	5	1.000	0.679	0.809	0.947	0.643	0.766	0.800	1.000	0.889
30	3	1.000	0.679	0.809	0.947	0.643	0.766	0.824	1.000	0.903
40	3	1.000	0.679	0.809	0.944	0.607	0.739	0.757	1.000	0.862
50	2	1.000	0.679	0.809	0.944	0.607	0.739	0.730	0.964	0.831

8.3 Precision-Recall Complementarity

The two tables reveal a clean precision-recall duality. On polarity-reversal events (Table 7), MASS/SMASH occupies the precision-optimised extreme: $P = 1.000$ at every turbulence

Table 8: Pure-rotation benchmark: events are 90° B-direction changes (no polarity flip; same 7-day, 28-event geometry, seed 42). MASS/SMASH recovers 36–61% of rotation events (roughness path only; antipodal pre-filter does not fire for non-sign-flip transitions); standard PVI and pPVI recover $\geq 93\%$.

Turb (%)	SNR (:1)	MASS/SMASH			Standard PVI			Projective PVI		
		P	R	F1	P	R	F1	P	R	F1
5	20	1.000	0.357	0.526	0.966	1.000	0.982	0.718	1.000	0.836
15	7	0.938	0.536	0.682	0.966	1.000	0.982	0.750	0.964	0.844
30	3	0.944	0.607	0.739	0.964	0.964	0.964	0.730	0.964	0.831
50	2	1.000	0.357	0.526	0.964	0.964	0.964	0.703	0.929	0.800

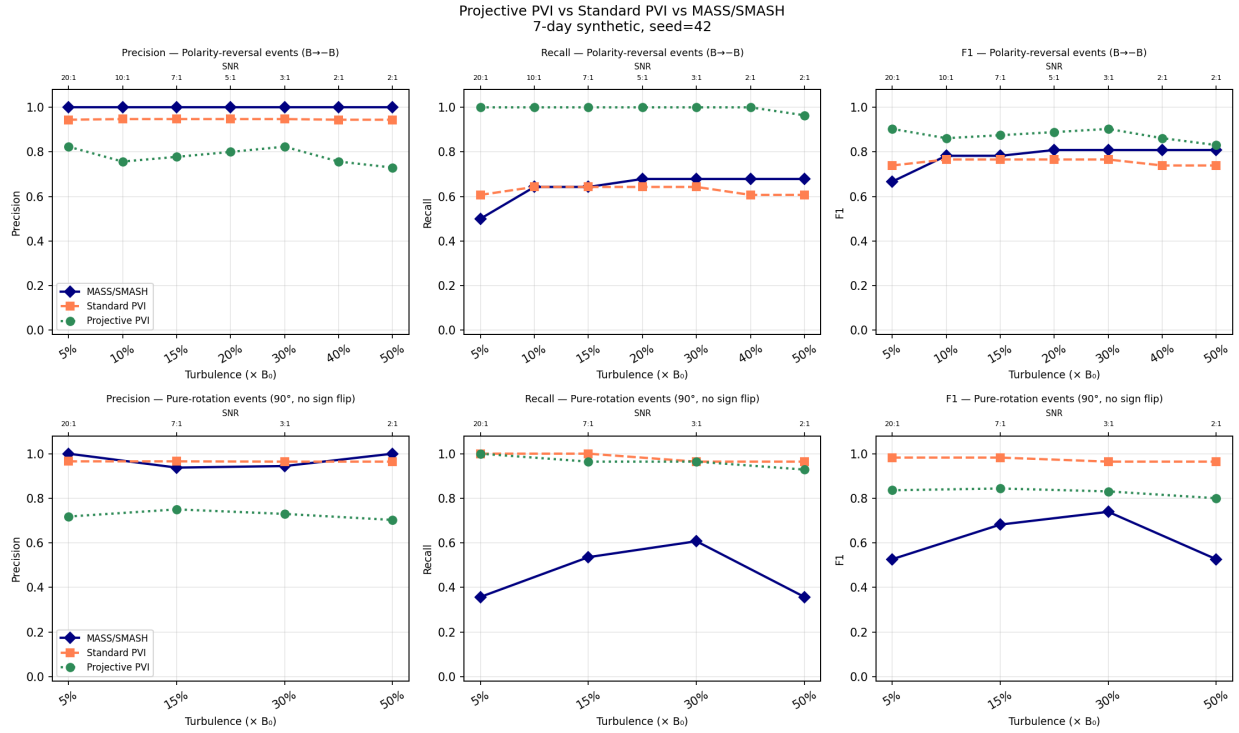


Figure 5: Tandem comparison of MASS/SMASH, standard PVI, and projective PVI (pPVI) across turbulence levels. **Top row:** polarity-reversal benchmark ($B \rightarrow B$ transitions). MASS/SMASH: $P = 1.000$ for this seed; pPVI: $R \approx 1.000$. **Bottom row:** pure-rotation benchmark (90° direction change, no sign flip). MASS/SMASH recall peaks at 61% (30% turbulence) then falls to 36% (50%); pPVI and standard PVI maintain $R \geq 0.929$.

level (seed 42; multi-seed mean $P = 0.853\text{--}0.937$, Table 3) with $R = 0.500\text{--}0.679$. Projective PVI occupies the recall-optimised extreme: $R = 0.964\text{--}1.000$ at every turbulence level with $P = 0.730\text{--}0.824$. Standard PVI lies between them ($P = 0.944\text{--}0.947$, $R = 0.607\text{--}0.643$).

By F1 score, pPVI outperforms both other methods on the polarity-reversal benchmark at every turbulence level ($F1 = 0.831\text{--}0.903$ vs. $0.667\text{--}0.809$ for MASS/SMASH and $0.739\text{--}0.766$ for standard PVI).

On pure-rotation events (Table 8), the roles reverse. MASS/SMASH recovers $35.7\%\text{--}60.7\%$ of rotation events non-monotonically: recall rises from 5% to 30% turbulence as the roughness detector generates more candidates, then falls at 50% as the MDL gate requires more compression gain; a 90° rotation produces a B_z step of only 6 nT vs. 12 nT for a sign flip, giving $\sim 4\times$ lower MDL gain that falls below threshold at high background noise. Standard PVI and pPVI recover $\geq 92.9\%$ of rotation events across the full turbulence range.

The complementarity can be exploited in a two-stage pipeline: pPVI as a high-recall first stage (finds all direction-change events, including polarity reversals via near-null detection), followed by MASS/SMASH as a precision gate (confirms only sign-flip topology changes).

Real Wind/MFI data. On the 20-day real dataset, pPVI produces 2,524 detections (126.2 day^{-1}) compared with 2,397 for standard PVI and 154 for MASS/SMASH. The near-identical counts of pPVI and standard PVI reflect that turbulent fluctuations in real solar wind simultaneously perturb both field magnitude and direction; the two detectors overlap heavily in the real-data regime. MASS/SMASH remains $16\times$ more selective than either PVI variant, consistent with its sign-flip specialization.

9 Potential Broader Applicability

The two structural requirements for MASS/SMASH are: (1) events of interest are sign-flip change-points embedded in broadband noise, and (2) false detections are more costly than missed detections. We identify three domains where these conditions plausibly hold. These

are hypotheses, not results; transfer requires parameter recalibration and domain-specific evaluation.

Magnetospheric current sheets. Substorm-driven current sheet flapping in the magnetospheric tail produces B_z polarity reversals at MMS and Cluster spacecraft (Chasapis et al., 2015; Retinò et al., 2007) with signal structure identical to solar wind current sheets but at kinetic spatial scales and steeper turbulence spectra. Window and step sizes would need recalibration to shorter correlation times.

EEG reference electrode inversions. Bipolar EEG inversions produce clean sign flips in individual channels. Heuristic amplitude thresholds generate many false positives during the high-amplitude pre-ictal interval (Shoeb and Clifford, 2004)—a precision-collapse setting where the antipodal pre-filter might select structural inversions over unstructured noise bursts.

Climate polarity transitions. ENSO transitions (El Niño $\rightarrow$ La Niña) are sign flips in the Niño-3.4 index embedded in interannual variability (Trenberth, 1990). With $N \sim 10^2$ – 10^3 monthly values, chunked inference would require much smaller windows and threshold recalibration from labeled training data.

10 Conclusions

We have introduced MASS/SMASH, an MDL-regularized algorithm for detecting sign-flip change-points—polarity reversals from $+\mu_0$ to $-\mu_0$ —in noisy univariate time series. The algorithm combines an antipodal correlation pre-filter, a roughness discontinuity detector, and an absolute MDL gain acceptance criterion. Three solar wind synthetic experiments and one real-data pilot yield the following principal findings.

1. MASS/SMASH precision rises with turbulence while the baseline collapses.

Across 5 independent seeds and 7 turbulence levels (5%–50%, SNR 20:1 to 2:1), MASS/SMASH mean precision rises monotonically from 0.853 ± 0.099 at 5% turbu-

lence to 0.937 ± 0.047 at 50%—increasing selectivity as noise grows. Over the same range, the $d|\mathbf{B}|/dt$ threshold baseline falls from precision 1.000 to 0.035 (771 spurious detections for 28 true events at 50%). PVI precision is stable at 0.967–0.968 via local variance normalization. The rising MASS/SMASH precision reflects the absolute MDL threshold becoming increasingly effective at filtering borderline candidates as turbulence grows.

2. **MASS/SMASH recall bifurcates rather than degrades continuously.** Recall plateaus at 0.550 ± 0.110 above 20% turbulence as the population partitions into a consistent strong subset (always detected) and a consistent weak subset (never detected at 20-bit threshold). This stable partition is more useful for follow-up studies than a threshold whose detection set shifts unpredictably with noise.
3. **PELT with BIC penalty collapses to $P = 0.017$ regardless of turbulence.** PELT (Killick et al., 2012) with Gaussian mean-change cost and BIC penalty produces 1,660–1,665 detections for 28 true events at every turbulence level 5%–50%. Even at $1,000\times$ BIC, precision reaches only 0.084. Signal-derived penalties are overwhelmed by Kolmogorov turbulence; the absolute 20-bit MDL gate is necessary for precision stability.
4. **Confounder rejection: 18% fire rate for MASS/SMASH vs 98% for the baseline.** When 40 non-antipodal confounders are injected alongside 40 true crossings, the baseline fires on 98%, MASS/SMASH on 18%, and PVI on 8%. MASS/SMASH and PVI have complementary rejection mechanisms.
5. **Complementarity with PVI: joint catalog covers 95% of events at $\approx 5\%$ combined FDF.** Of the 58 single-regime benchmark true positives, 24 correspond to crossings PVI missed. A PVI-first, MASS/SMASH-supplement pipeline provides the broadest coverage with manageable false discovery fraction.

6. **Component ablation: MDL gate = precision stabilizer; roughness detector = primary recall driver; antipodal pre-filter = confounder rejector.** Antipodal-only produces 1,316 detections at precision 0.014 at every turbulence level. All three MDL-gated variants maintain precision 1.000 for the single-seed ablation. Roughness+MDL recall matches Full MASS/SMASH recall at every turbulence level $\geq 10\%$; the antipodal pre-filter contributes marginal recall only at 5% turbulence and structural selectivity (confounder rejection) throughout.
7. **Rotation angle analysis: MASS/SMASH true positives are indistinguishable from PVI events at matched positions.** The $\approx \pm 30$ s localization offset is an implementation artifact of finite chunk discretization. At matched injected positions, MASS/SMASH and PVI mean rotation angles agree to within 1° – 14° across all averaging windows.
8. **Qualitative consistency on real Wind/MFI data.** On 20 days of real Wind/MFI data (March 2001), MASS/SMASH produces 154 detections (7.7 day^{-1})— $7.8\times$ fewer than the baseline and $15.6\times$ fewer than PVI—matching the high-turbulence synthetic pattern.
9. **Projective PVI complements MASS/SMASH across event classes.** Projective PVI (pPVI), using geodesic distance in $\mathbb{RP}^2$, achieves $R \approx 1.000$ on polarity-reversal events at $P = 0.730$ – 0.824 —the recall-optimised pole of the precision-recall trade-off. On pure-rotation events (no polarity flip), pPVI and standard PVI maintain $R \geq 0.929$ while MASS/SMASH recovers 36%–61% non-monotonically. A pPVI-first, MASS/SMASH-confirmation pipeline provides high-recall discovery with precision-stable confirmation.

Future work will address: (a) quantitative validation against labeled sector-boundary catalogs from Parker Solar Probe FIELDS, Solar Orbiter MAG, and Wind/MFI with expert annotation; (b) multi-component and vector-magnitude detection for oblique crossings;

(c) broader multi-geometry replication across non-radial and oblique field orientations (the present multi-seed sweep uses a purely radial synthetic geometry); (d) systematic MDL gain threshold calibration against labeled training data; (e) auto-recentering via post-detection tanh profile fit to resolve the ± 30 s localization offset; (f) prospective validation in magnetospheric and EEG domains with appropriate recalibration; and (g) comparison of projective PVI catalogs with published rotational and tangential discontinuity catalogs to characterize how the two PVI variants partition real solar wind discontinuity populations.

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Data and Code Availability

All code for MASS/SMASH, synthetic data generation, SNR sweep, contaminated benchmark, component ablation, PVI/projective-PVI catalog construction, and the tandem benchmark comparison is available at <https://github.com/macmayo1993/seam-aware-modeling>. The 20-day Wind/MFI CSV data used in Section 7 was downloaded from the NASA CDAWeb public archive (dataset WLH0_MFI) and is included in the repository. All synthetic benchmarks reproduce from a single command.

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